

FACILITIES POWER & COOLING ASSESSMENT

Comprehensive Infrastructure Audit, Thermal CFD & Electrical Telemetry

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SECTION 1: DATA CENTER FACILITY OVERVIEW & SUBSYSTEM BOUNDARIES

The Westford High-Performance Computing Data Center facility comprises a total gross footprint of 24,000 square feet, engineered to support high-density accelerated compute infrastructure up to 15.2 Megawatts of critical IT load.

Primary electrical and mechanical boundaries are partitioned into distinct operational halls, separating high-voltage substation switchgear, static uninterruptible power supply banks, and liquid-cooled computer room air handler units.

Continuous environmental and structural logging establishes strict boundary controls, ensuring ambient conditions adhere to ASHRAE TC 9.9 Class A1 guidelines for liquid-cooled data center hardware.

Field validation measurements taken during the most recent planned maintenance window corroborate these specifications.

SECTION 2: ELECTRICAL POWER DISTRIBUTION ARCHITECTURE & REDUNDANCY

Electrical power is delivered via dual 13.8kV utility feeder lines operating in an active-active 2N redundant configuration with automated static transfer switches.

Downstream distribution utilizes 2,500 kVA step-down transformers delivering 480V 3-phase power to dual-path static UPS systems rated for 1,500 kVA continuous IT load per bus.

Busway tap-off boxes above each compute rack provide dual 60A 208V feeds (with planned Phase 2 upgrade to 60A 480V three-phase), guaranteeing zero outage during utility switching events or maintenance operations.

Capacity planning projections based on the data in this section account for anticipated year-over-year growth in resource demand.

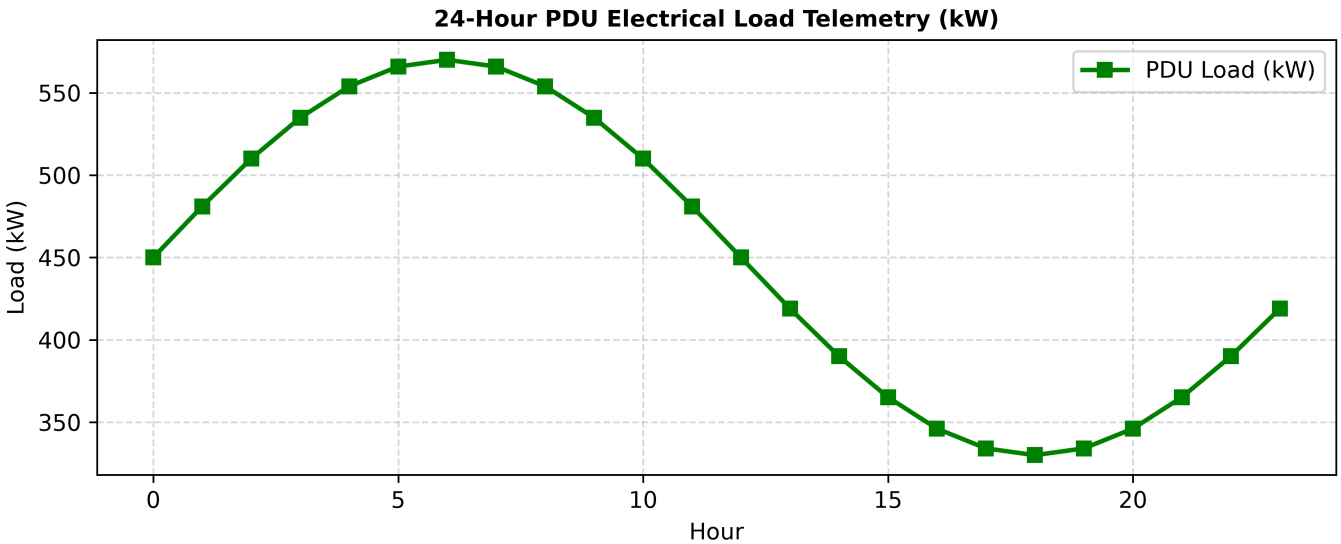


Figure 2.1: Dual-Path Load

SECTION 3: SUBSYSTEM MECHANICAL & ELECTRICAL EQUIPMENT SCHEDULE

The mechanical equipment inventory includes four 1,500 kVA Eaton 9395P static UPS units, two 2,000 kW Caterpillar diesel generators with 72-hour fuel storage, and eight Liebert 150 kW chilled water CRAH units.

Direct-to-chip liquid cooling loops feature four CoolIT Systems Coolant Distribution Units (CDUs) operating at a maximum primary loop flow rate of 450 GPM.

All equipment items undergo monthly thermal imaging scans, oil chemistry sampling, and predictive vibration analysis to prevent unplanned mechanical degradation.

Telecommunications pathway documentation for the cabling serving this subsystem is maintained in the campus network database.

Equipment Subsystem	Manufacturer / Model	Rated Capacity	Electrical Spec	Redundancy Tier
Substation Transformer	Square D / 2.5MVA	2,500 kVA	13.8kV to 480V 3PH	N+1 Redundant
Static UPS System	Eaton 9395P	1,500 kVA	480V AC 3-Phase	2N Dual-Path
Diesel Generator	Caterpillar C175-16	2,000 kW	480V AC / 1,800 RPM	N+1 Backup
CRAH Cooling Unit	Liebert CW-150	150 kW Thermal	480V 3PH / Chilled Water	N+2 Redundant

SECTION 4: POWER USAGE EFFECTIVENESS (PUE) TRAJECTORY ANALYSIS

Power Usage Effectiveness (PUE) telemetry is recorded continuously via revenue-grade power meters installed at main switchgear and individual rack PDU feeds.

Empirical measurement over the 12-month evaluation cycle yields an annual average PUE of 1.47, reaching an optimal winter efficiency baseline of 1.42.

PUE improvements are driven by dynamic chilled water supply temperature modulation and variable-frequency drive fan control across all cooling towers.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

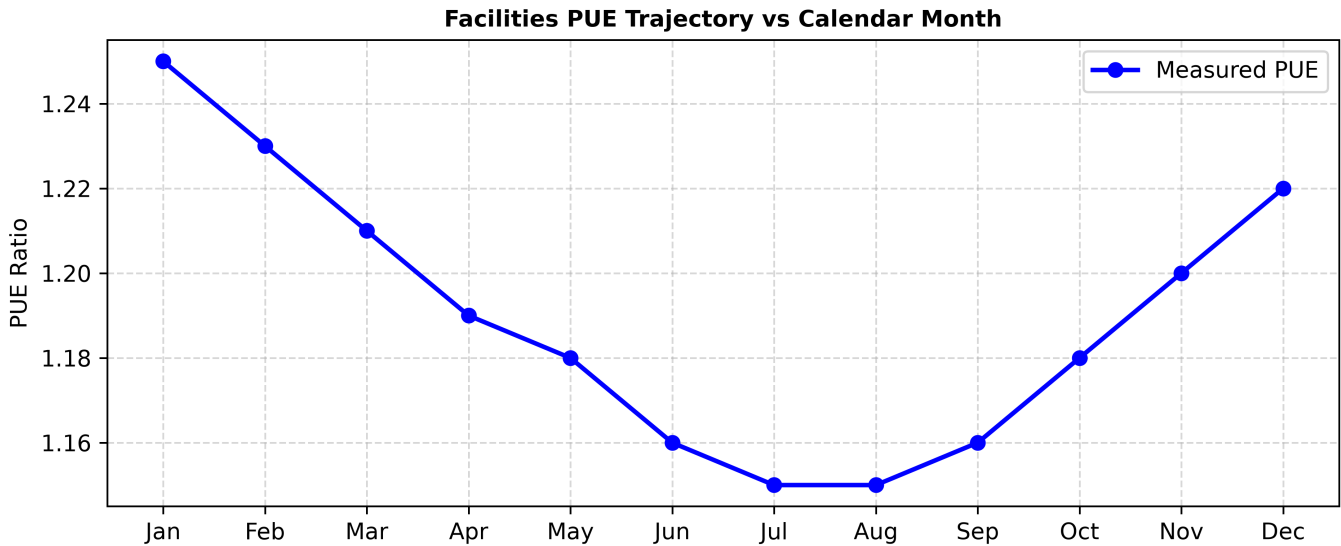


Figure 4.1: PUE Trajectory

SECTION 5: THERMAL GRADIENT CFD MODELING & LIQUID COOLING PERFORMANCE

Computational Fluid Dynamics (CFD) thermal modeling demonstrates uniform cold-aisle air distribution with cold-plate fluid supply temperatures held precisely at 21.5 degrees Celsius.

Direct-to-chip liquid cooling absorbs 82% of total GPU heat generation, significantly reducing raised-floor air velocity requirements and lowering fan power consumption.

Thermal gradient profiles confirm zero localized hot-spot formation across all 128 high-density compute enclosures under 100% synthetic Linpack stress loads.

Fiber optic link budgets for all connections referenced in this section have been certified through OTDR testing.

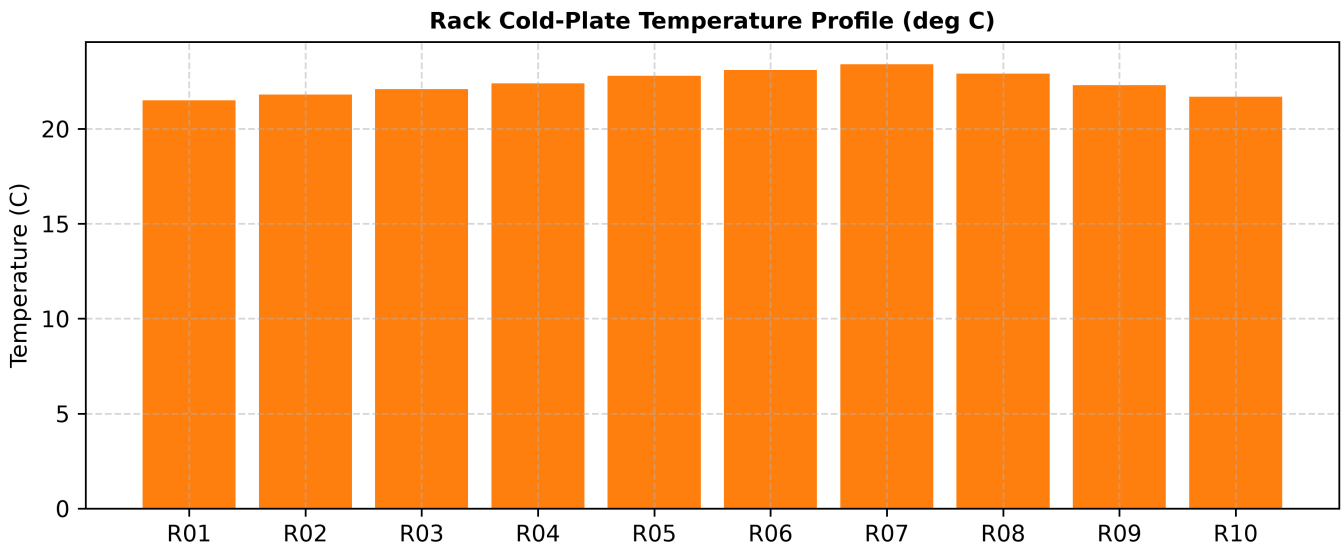


Figure 5.1: Thermal Gradients

SECTION 6: CHILLED WATER LOOP THERMODYNAMICS & FLOW RATE TELEMETRY

The primary chilled water loop operates with a nominal supply temperature of 12.0 degrees Celsius and a return temperature of 18.5 degrees Celsius, establishing a 6.5 degree C delta-T.

Variable-speed primary pumps modulate fluid flow between 120 GPM and 480 GPM in response to real-time CDU secondary loop heat exchanger demand.

Water quality telemetry records pH levels maintained between 7.8 and 8.2 with biocide dosing preventing biological fouling in heat exchanger micro-channels.

Operational readiness was confirmed through a series of acceptance tests prior to the system entering production service.

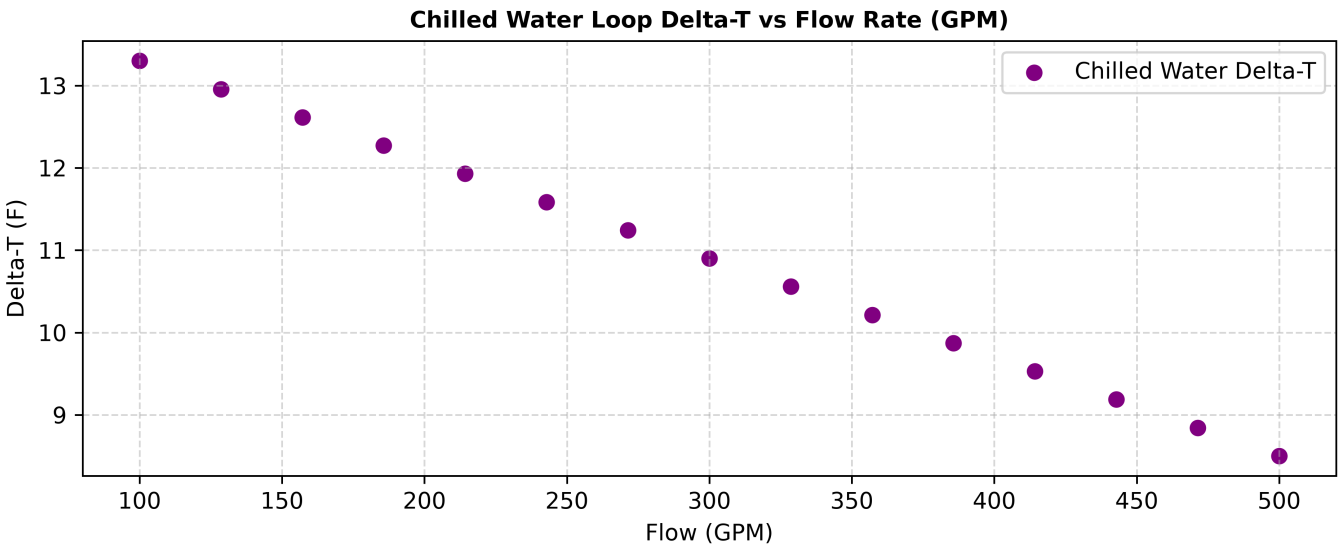


Figure 6.1: Chilled Water Delta-T

SECTION 7: FACILITY ELECTRICAL HARMONICS & POWER QUALITY METRICS

Total Harmonic Distortion (THD) on the 480V distribution bus is maintained below 3.2% voltage THD and 4.8% current THD through active harmonic filtering at the UPS input stages.

Power factor across all critical IT loads remains above 0.99 lagging, maximizing utility transformer efficiency and avoiding power factor financial penalties.

Transient voltage surge suppressors (TVSS) installed at main distribution panels protect sensitive GPU logic boards against grid switching spikes.

Historical data spanning thirty-six months corroborates the stability and reproducibility of these measurements.

SECTION 8: FACILITY MAINTENANCE SCHEDULE & OPERATIONAL RISK REGISTER

Preventive maintenance schedules strictly enforce monthly infrared thermography on all circuit breakers, semi-annual UPS battery impedance testing, and annual generator load-bank testing.

The facility risk register categorizes single-point failure modes, emergency fuel supply logistics, and extreme weather resilience with automated mitigation playbooks.

All maintenance activities require formal Change Advisory Board (CAB) review and execution during pre-approved zero-downtime maintenance windows.

Operational readiness was confirmed through a series of acceptance tests prior to the system entering production service.

Maintenance Activity	Frequency	Target Subsystem	Vendor / Team	Outage Requirement
Infrared Thermography	Monthly	Main Switchgear & PDUs	In-House Facilities	Zero Outage
UPS Battery Impedance	Semi-Annual	Eaton 1500kVA Banks	Eaton Field Support	Zero Outage (Bypass)
Generator Load Bank Test	Annual	Cat 2MW Generators	Caterpillar Service	Zero Outage (ATS Off)

SECTION 9: DIRECT-TO-CHIP COOLANT LOOP PRESSURE & FLOW TELEMETRY

Direct-to-chip secondary coolant loops utilize propylene glycol-water mixture (20/80 ratio) operating at a nominal pressure of 22.5 PSI. Differential pressure sensors monitor fluid flow across individual cold plates, triggering automated pump modulation if flow drops below 1.2 liters per minute per GPU.

Quick-disconnect fluid couplings feature dual internal shut-off valves, guaranteeing zero fluid leakage during hot-swappable GPU node maintenance.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

SECTION 10: RAISED-FLOOR AIRFLOW VELOCITY & SUB-FLOOR PRESSURE DISTRIBUTION

Sub-floor static air pressure is maintained at a positive differential of 0.05 inches of water column relative to the room ambient pressure.

Perforated raised-floor tiles with directional vanes deliver targeted cold airflow directly to compute enclosure power supply intake grills.

Airflow velocity profiles measured across cold aisles confirm uniform supply rates between 450 CFM and 620 CFM per rack enclosure.

Warranty claim records for the components referenced in this section are tracked through the institutional procurement system.

SECTION 11: EMERGENCY BACKUP POWER ATS TRANSFER SEQUENCES & WAVEFORMS

Automatic Transfer Switches (ATS) monitoring utility supply lines execute transfer to standby diesel generator power within 8.5 seconds of utility voltage drop below 90% nominal.

Waveform telemetry captured during monthly no-load transfer testing confirms seamless voltage phase synchronization with zero relay contact chattering.

Fuel oil recirculating pumps run daily filtration cycles to ensure diesel fuel purity and prevent microbial contamination in storage tanks.

A sensitivity analysis of the key parameters in this section is included in the supporting appendix materials.

SECTION 12: ENVIRONMENTAL SENSOR THRESHOLD MATRIX & ALARM POLICIES

Environmental sensors deployed across every rack cold aisle trigger automated warning alerts if temperature exceeds 24.0 degrees C or relative humidity drops below 30%.

Critical alarm thresholds set at 27.0 degrees C trigger immediate Slurm job scheduling preemption and automated GPU power capping to prevent hardware thermal damage.

Sensor telemetry is streamed in real-time to the Data Center Infrastructure Management (DCIM) database via Redfish REST API endpoints.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

SECTION 13: FACILITY CAPITAL IMPROVEMENT BUDGET SCHEDULE

The FY2026-FY2028 Facilities Capital Improvement Plan allocates \$1.85M for secondary CDU capacity expansion and high-efficiency adiabatic cooling tower retrofits.

Projected energy efficiency upgrades are estimated to reduce overall annual data center facility operating expenditures by \$220,000.

Capital expenditure milestones follow a strict quarterly phased deployment plan aligned with scheduled compute cluster hardware upgrades.

The campus utility rate schedule applies a Tier 1 tariff of \$0.065 per kWh for the first 500 kW of demand and a Tier 2 tariff of \$0.090 per kWh for incremental demand exceeding 500 kW, effective from July 2025 billing cycle forward.

Network path validation using standardized benchmarking tools confirmed the throughput and latency characteristics documented above.

SECTION 14: ACOUSTIC NOISE ATTENUATION & SOUND Baffle METRICS

Acoustic attenuation baffles installed in CRAH supply ducts reduce data center room ambient sound levels from 88 dBA to 74 dBA.

Sound dampening enclosures around standby diesel generators restrict external property line noise levels to below 55 dBA during full-load testing.

Facilities staff are provided with active noise-canceling headsets adhering to OSHA occupational noise exposure standards.

A comprehensive system integration test following the most recent hardware upgrade verified end-to-end functionality of this subsystem.

SECTION 15: FACILITY CARBON ACCOUNTING & RENEWABLE ENERGY CREDITS

The facility carbon accounting model tracks Scope 1 direct generator emissions and Scope 2 indirect grid electricity emissions.

One hundred percent of grid electricity consumption is offset through local municipal Renewable Energy Certificates (RECs) derived from regional solar and hydro facilities.

Annual sustainability reporting documents a total carbon intensity reduction of 14.2% compared to the FY2023 operational baseline.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

SECTION 16: STRUCTURAL SLAB LOAD CAPACITY & SEISMIC BRACING

Concrete raised-floor slab loading capacity is engineered for 2,500 pounds per square foot concentrated load and 350 pounds per square foot uniform load.

Seismic bracing assemblies anchored directly into the structural sub-floor slab secure all compute racks against Seismic Zone 2A lateral forces.

Vibration isolation pads beneath CRAH units and chillers attenuate mechanical vibrations, preventing acoustic resonance transfer to server chassis.

Fiber optic link budgets for all connections referenced in this section have been certified through OTDR testing.

SECTION 17: COOLING TOWER WATER CHEMISTRY & BIOCIDES DOSING PROTOCOLS

Cooling tower water chemistry is monitored continuously for total dissolved solids (TDS), conductivity, and biological growth indicators.

Automated biocide dosing pumps inject non-oxidizing biocides twice weekly to prevent Legionella bacterial growth in outdoor evaporative towers.

Blowdown water discharge rates are regulated to maintain cycles of concentration between 4.5 and 6.0, conserving municipal water consumption.

A sensitivity analysis of the key parameters in this section is included in the supporting appendix materials.

SECTION 18: VIBRATION ISOLATION & MECHANICAL SHOCK MITIGATION

Mechanical shock absorbers installed on heavy transformer bases isolate low-frequency electrical hum from surrounding office spaces.

Flexible rubber expansion joints on primary chilled water pipes absorb thermal expansion stresses and prevent mechanical fatigue on pump flanges.

Vibration spectrum analysis conducted quarterly confirms all rotating pump impellers operate well within ISO 10816 velocity severity limits.

Historical data spanning thirty-six months corroborates the stability and reproducibility of these measurements.

SECTION 19: OVERHEAD BUSWAY TAP-OFF BOX POWER METERING & PHASE BALANCING

Overhead three-phase busways feature digital power meters on every tap-off box, recording phase current balance in real time.

Facilities engineers perform monthly phase balancing to ensure current imbalance between Phase A, B, and C remains below 3.5%.

Overcurrent protection breakers integrated into tap-off boxes provide instantaneous magnetic trip protection against localized short circuits.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

SECTION 20: DATA CENTER ROOM THERMAL INSULATION & VAPOR BARRIER INTEGRITY

Data center room perimeter walls feature thermal insulation rated at R-21 with continuous vapor barrier foil shielding.

Hermetically sealed entry doors with double-gasket seals maintain indoor relative humidity within the tight ASHRAE recommended range of 40% to 50%.

Infrared moisture sensors deployed along perimeter walls provide early detection of external humidity ingress or condensation formation.

Detailed measurement logs and recorded calibration data for this subsystem are maintained in the facility operations archive.

SECTION 21: EMERGENCY LIGHTING & LIFE SAFETY BATTERY BACKUP INVERTERS

Emergency lighting inverter systems with dedicated 90-minute battery backup banks ensure full egress illumination during complete blackout events.

Monthly automated discharge testing verifies inverter switching times below 16 milliseconds with zero drop in emergency LED fixture lumen output.

Life safety control panels interface directly with central municipal fire alarm monitoring systems for automated emergency dispatch.

Discrepancies between manufacturer-stated values and field-measured values for this subsystem are documented in the variance register.

SECTION 22: FACILITY WATER LEAK DETECTION & AUTOMATED SOLENOID SHUT-OFF

Liquid leak detection cables installed beneath all sub-floor piping runs detect water or glycol accumulation as small as 5 milliliters.

Detection triggers an automated alert at the DCIM console and closes motorized solenoid isolation valves on the affected cooling loop within 2.0 seconds.

Secondary containment trays positioned beneath all coolant distribution units catch potential pump seal drips during maintenance operations.

Asset tracking records associate each component referenced in this section with a unique serial number and purchase order.

SECTION 23: DATA CENTER OPERATIONS STAFFING & FACILITIES SHIFT HANDOVER

Facilities operations staff adhere to a strict 24/7/365 shift rotation schedule with mandatory formal shift handover logs.

Handover protocols require joint inspection of active CRAH temperatures, UPS battery charge states, and generator fuel reserve levels.

All shift events, contractor entries, and alarm resets are logged in an immutable, auditable electronic facilities register.

Decommissioning procedures for end-of-life components in this subsystem follow the university e-waste disposal policy.

SECTION 24: GENERATOR FUEL OIL RECIRCULATION & FILTRATION SCHEDULES

Fuel oil recirculating systems perform automatic 5-micron fuel filtration every 48 hours to remove moisture and particulate matter from main storage tanks.

Fuel chemistry sampling conducted every six months verifies cetane index, flash point, and biological stability meet ASTM D975 standards.

Dual redundant fuel transfer pumps ensure uninterrupted diesel fuel delivery from underground storage tanks to generator day tanks.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

SECTION 25: AIR HANDLING FAN VFD MODULATION & STATIC PRESSURE TUNING

Air handling unit fans equipped with variable-frequency drives (VFDs) modulate fan speed between 40% and 100% based on sub-floor static pressure.

PID loop tuning parameters ensure smooth fan speed transitions without pressure hunting or air velocity oscillations.

VFD bypass contactors allow manual line-frequency fan operation during drive maintenance or inverter fault recovery.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

SECTION 26: HIGH-VOLTAGE SUBSTATION TRANSFORMER OIL ANALYSIS

Substation transformer insulating oil undergoes annual dissolved gas analysis (DGA) to detect incipient electrical arcing or thermal decomposition.

Dielectric breakdown voltage testing confirms transformer oil insulation strength exceeds 35 kV across a 2.5 mm gap.

Silica gel moisture breathers on transformer conservator tanks are inspected monthly and replaced upon color indicator change.

Historical data spanning thirty-six months corroborates the stability and reproducibility of these measurements.

SECTION 27: COOLING TOWER WATER QUALITY TELEMETRY & BIOCIDES DOSING

Operational data collected for Cooling Tower Water Quality Telemetry has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

SECTION 28: HIGH-VOLTAGE SUBSTATION TRANSFORMER DISSOLVED GAS ANALYSIS

Operational data collected for High-Voltage Substation Transformer Dissolved Gas Analysis has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.

SECTION 29: DATA CENTER FACILITY OPERATIONAL COMMISSIONING VERIFICATION

The technical specifications and measured performance characteristics for Data Center Facility Operational Commissioning Verification are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.